

KEY

MATHEMATICS Introduction to Modern Algebra I Quiz 2.

1) Let $f: S \rightarrow T$ and $g: T \rightarrow U$ be functions. Prove that if $g \circ f$ is surjective, then:

- (1) g is surjective.
- (2) f may not be surjective.

1) If $u \in U$, then since $g \circ f$ is onto, $u = g(f(s))$ for some $s \in S$. But then $u = g(t)$ where $t \in T$ ~~and~~ $t = f(s)$

2) Let $S = U = \mathbb{N}$, $T = \mathbb{Z} - \{0\}$, $f(s) = s$, $g(t) = |t|$

Then $(g \circ f)(s) = |s| = s$ is onto but f is not onto because -1 is not image of f .

2) If $f \in S_n$, show that $f^k = i$ for some integer $1 \leq k \leq n!$

Pt Let $S_n = A(1, \dots, n)$. We have $f^j(n) = f^l(n)$ for some $0 \leq j < l \leq n$, so applying f^{-j} , we have $n = f^{l-j}(n)$ where $0 < l-j \leq n$. So we get $g: \{1, 2, \dots, n-1\} \rightarrow \{1, 2, \dots, n-1\}$ by letting $g(i) = f^{l-j}(i)$

By induction $g^{k'} = \text{id}$ for $1 \leq k' \leq (n-1)!$

Thus $f^{k'(l-j)} = \text{id}$. Let $k = k'(l-j)$ then $1 \leq k \leq n(n-1)! = n!$

2nd Pt $|S_n| = n!$ so $f^i = f^l$ some $0 \leq i < l \leq n!$

Applying f^{-i} we get $\text{id} = f^{l-i}$ some $0 < l-i \leq n!$